LIAM COULTER

liamjcoulter@gmail.com ⋄ U.S. Citizen ⋄ https://liamjcoulter.github.io

WORK EXPERIENCE

RTX BBN Technologies St. Louis Park, Minnesota *August 2023 - Present*
Sr. Engineer (Research), Spectrum Dominance group

- Regularly brief technical results and progress to end customer including program kickoffs, quarterly progress updates, and final reports
- Prioritize, plan and execute technical work, coordinate team members, and balance near-term priorities with long-term strategic planning to benefit wider team and suite of programs
- Participate in multiple field demonstrations of experimental technology to end customer including preparation & planning, on-site troubleshooting, and direct customer interaction, at government & RTX sites
- Contribute to project proposals and presentations for new & existing external customers to win new work and maintain strong relationships
- Spectrum Dominance group focused on RF, signal processing, and signals intelligence, within Non-kinetic Capabilities and Technologies business unit. Concurrently pursuing PhD

RTX BBN Technologies St. Louis Park, Minnesota *May 2020 - August 2023*
Staff Scientist I, Spectrum Dominance group

- Drive program execution for a wide range of government customers, as well as Raytheon internal research & development projects
- Develop & implement algorithms for signal and image processing across a wide range of technical areas including passive & remote sensing, distributed systems, radar signal processing, geolocation & target tracking, and electromagnetic propagation modeling

NASA Goddard Space Flight Center Greenbelt, Maryland *August 2019 - May 2020*
Graduate Student Researcher, Supervisor: Dr. Joel McCorkel

- Developed experimental data processing pipeline and Python implementation of radio occultation using geostationary satellite, with real NASA data from GOES-16/GOES-R spacecraft
- Presented results at American Meteorological Society Fifth Conference on Atmospheric Biogeosciences, June 2021 (abstract published), and received "Outstanding Oral Presentation" award

Autonomy Technology Research Center Dayton, Ohio *May 2019 - August 2019*
Graduate Student Research Intern, Supervisor: Dr. Chad Waddington

- Implemented and compared algorithms for passive radar emitter localization (Python & MATLAB)
- Utilized high-performance computing clusters for simulation experiments, presented results to senior Air Force Research Laboratory scientists

EDUCATION

University of Minnesota Minneapolis, Minnesota *September 2020 - Present (Expected Fall 2026)*
PhD Candidate, Electrical and Computer Engineering. Advisor: Dr. Jarvis Haupt GPA 3.88

- Developing novel approaches to imaging without a line-of-sight, and investigating fundamental bounds for recovering data from light which has undergone diffuse reflection
- Classwork: estimation & detection theory, compressed sensing, stochastic processes, image processing, parallel computing, and data compression
- PhD candidate, planned graduation 2026. Concurrently working (see above)

Colorado State University Fort Collins, Colorado *August 2017 - August 2019*
MSc., Mathematics, Advisor: Dr. Margaret Cheney GPA 3.54

- Master's thesis: "Numerical Simulations for Synthetic Aperture Source Localization"

University of St. Thomas St. Paul Minnesota *September 2012 - May 2017*
BA, Mathematics (Minors Mech. Eng. & Philosophy) GPA 3.76

PUBLICATIONS

Preprints (Available upon request)

- **Liam J. Coulter** and M. A. Bliss, "Resolving Ambiguity in Passive TDOA Geolocation Systems with an Altitude-Parametrized EKF," **accepted to 2026** IEEE Asilomar Conf. on Sig., Systems, & Computers.
- **Liam J. Coulter**, A. V. Sambasivan, R. G. Paxman, and Jarvis D. Haupt, "Computational Information-Theoretic Analysis for Parameter Estimation in Optical Imaging Systems," *IEEE Trans. Comput. Imaging*, **submitted May, 2026**.
- A. V. Sambasivan, **Liam J. Coulter**, R. G. Paxman, and Jarvis D. Haupt, "On the Effects of Computational Forward Model Uncertainty in Parameter Estimation Lower Bounds," *IEEE Trans. Inf. Theory* **submitted June, 2026**.

Refereed Papers

- **Liam J. Coulter**, D. J. Geroski and A. C. Marcum, "Joint Recursive Receiver and Emitter Geolocation Using Time Difference of Arrival Measurements", **2023** 57th IEEE Asilomar Conf. on Sig., Systems, and Computers, USA, pp. 1277-1282, doi: 10.1109/IEEECONF59524.2023.10476880.
- S. Gleason, I. Cherniak, I. Zakharenkova, D. Hunt, S. Sokolovskiy, D. Freesland, A. Krimchansky, J. McCorkel, **Liam Coulter**, G. Ramsey, and J. Chapel, "The First Atmospheric Radio Occultation Profiles From a GPS Receiver in Geostationary Orbit", IEEE Geosci. Remote Sens. Lett., vol. 19, pp. 1-5, **2022**, Art no. 1005605, doi: 10.1109/LGRS.2022.3185828

Abstracts & Presentations

- **Liam Coulter**, J. McCorkel, and S. Gleason "Radio Occultation Measurements of Earth's Atmosphere from Geostationary Orbit for Weather Prediction and Water Vapor and Precipitation Monitoring," AMS 5th Conf. on Atmos. Biogeosciences, June **2021**. Received "Outstanding Oral Presentation" award.
- **Liam Coulter** and Margaret Cheney, "Synthetic Aperture Source Localization," SIAM Front Range Applied Mathematics (FRAM) graduate student conference presentation, March **2019**. Abstract published.

TECHNICAL SKILLS

Mathematics	Modeling · Probability & Statistics · Stochastic Processes · Linear Algebra · Optimization Inverse Problems · Numerical & Monte Carlo Methods · Statistical Inference
Signal Processing	Estimation & Detection · Cramér-Rao Bounds & Fisher Information · Image Processing Distributed & Remote Sensing · Kalman Filter · Radar Signal Processing · Sensor Fusion Time-Frequency Analysis · Cyclostationary Signal Processing · Electromagnetic Propagation
Software	MATLAB · Python · C/C++ & CUDA · Mathematica · Git · Linux · Parallel Computing Blender · Mitsuba renderer

LEADERSHIP & SERVICE

Founded Monthly Journal Club RTX BBN Technologies, Minnesota *July 2026 - Present*

- Initiated and launched monthly paper presentation/journal club for on-site technical staff
- Cultivate inclusive and collaborative environment, uncover potential areas for collaboration across teams
- Organize in-person meeting for local & remote staff, including managing logistics, securing presenters, and funding